

5.7.25. A certain computer algorithm executes twice as many operations when it is run with an input of size k as when it is run with an input of size $k - 1$ (where k is an integer that is greater than 1). When the algorithm is run with an input of size 1, it executes seven operations. How many operations does it execute when it is run with an input of size 25?

Suppose a certain computer algorithm executes twice as many operations when it is run with an input of size k , as when it is run with an input of size $k - 1$.

As in, k executes twice as many operations as $k - 1$, for some integer k , where $k > 1$.

Suppose the algorithm executes seven operations when run with an input of size 1.

We must show how many operations the algorithm executes when run with an input of size 25.

What do we know?

- $P(k)$ executes twice as many operations as $P(k - 1)$
- $P(1)$ executes seven operations
- k is some integer
- $k > 1$

a_k = number of operations for an input of size k

Since $P(k)$ executes twice as many operations as $P(k - 1)$, then:

$$a_k = 2a_{k-1}$$

And we know that $a_1 = 7$, hence:

$$a_2 = 2a_{2-1} = 2a_1 = 2 \cdot 7 = 14$$

$$a_3 = 2a_2 = 2 \cdot 14 = 28$$

$$a_4 = 2a_3 = 2 \cdot 28 = 56$$

From the recurrence relation:

$$a_k = 2a_{k-1}$$

We suppose to guess this pattern:

$$a_k = 7 \cdot 2^{k-1}$$

$$a_1 = 7$$

$$a_2 = 7 \cdot 2^1 = 14$$

$$a_3 = 7 \cdot 2^2 = 28$$

$$a_4 = 7 \cdot 2^3 = 56$$

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$$a_{25} = 7 \cdot 2^{24} = 117\,440\,512$$

Solution: The algorithm executes 117 440 512 operations with an input size 25.